

Equity Financing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Financing Restraint (EFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EFR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.93 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Financial markets process vast amounts of information daily, yet the speed and completeness of this information incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that markets exhibit systematic delays in processing certain types of signals, particularly those requiring careful analysis or interpretation (Hong et al., 2000; DellaVigna and Pollet, 2009). We examine this phenomenon through the lens of equity financing restraint (EFR), a signal that captures firms’ reluctance or inability to issue new equity despite potential growth opportunities. This measure provides a unique window into how markets gradually incorporate information about corporate financing constraints and their implications for future returns.

The slow diffusion of information provides a compelling framework for understanding why equity financing restraint predicts future stock returns. When firms exhibit restraint in equity issuance despite apparent needs or opportunities, this signal may not immediately capture investors’ attention, particularly for stocks with limited analyst coverage or lower institutional ownership (Hirshleifer and Teoh, 2003; Peng and Xiong, 2007). The gradual incorporation of this information creates predictable return patterns as the market slowly recognizes the implications of financing constraints for firm value and future cash flows (Hong et al., 2000).

Investor inattention and cognitive constraints fundamentally limit the speed at which information diffuses through markets. Research demonstrates that investors face bandwidth limitations that prevent them from simultaneously processing all available information, leading to underreaction to less salient signals (Kahneman, 1973; Hirshleifer et al., 2011). Equity financing restraint represents precisely such a signal—it requires synthesizing multiple pieces of information about a firm’s financial position, growth prospects, and market conditions, making it particularly susceptible

to delayed recognition (Cohen and Frazzini, 2008).

The concentration of return predictability among stocks with greater information frictions supports the gradual diffusion hypothesis. Small stocks, those with low analyst coverage, and firms with limited institutional ownership exhibit stronger return continuation following EFR signals because information travels more slowly in these segments (Hong et al., 2000; Zhang, 2006). This cross-sectional variation in return predictability directly links the EFR effect to underlying information processing mechanisms rather than risk-based explanations, as attention constraints bind more tightly where professional intermediaries are scarce (Hou and Moskowitz, 2009).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on equity financing restraint achieves an annualized gross Sharpe ratio of 0.53, with a monthly average excess return of 31 basis points (t -statistic = 4.15). The strategy’s performance remains robust after controlling for known risk factors, generating a monthly alpha of 22 basis points (t -statistic = 2.93) relative to the Fama-French six-factor model. These results establish EFR as an economically significant predictor of cross-sectional returns that cannot be explained by existing factor models.

The predictive power of equity financing restraint persists across various portfolio construction methodologies and after accounting for transaction costs. The strategy delivers a net Sharpe ratio of 0.47 and maintains statistical significance across all standard factor models, with net alphas ranging from 21 to 30 basis points per month. Importantly, when we control for the six most closely related anomalies from the factor zoo—including share issuance measures, growth in book equity, and net payout yield—the EFR strategy still generates a significant alpha of 15 basis points per month (t -statistic = 2.24), demonstrating that it captures unique information not reflected in existing predictors.

Consistent with the gradual information diffusion hypothesis, we find that return predictability concentrates among stocks where attention constraints are most bind-

ing. Among the largest stocks (those above the 80th NYSE percentile), the EFR strategy achieves an average return of 24 basis points per month (t-statistic = 2.68), while returns are substantially higher among smaller stocks where information frictions are more severe. The strategy’s gross Sharpe ratio ranks in the top 8% of 212 documented anomalies, and its net Sharpe ratio places it in the top 2%, highlighting both its economic significance and implementability after accounting for trading costs.

This paper makes three distinct contributions to the asset pricing literature on information diffusion and return predictability. First, we document that equity financing restraint represents a powerful yet underexplored signal that predicts cross-sectional returns through the channel of gradual information incorporation, extending the framework of [Hong et al. \(2000\)](#) and [DellaVigna and Pollet \(2009\)](#) to corporate financing decisions. Unlike mechanical accounting measures studied in prior work ([Cooper et al., 2008](#); [Richardson et al., 2005](#)), EFR captures a nuanced aspect of managerial decision-making that requires careful interpretation, making it particularly susceptible to delayed market recognition.

Second, our results demonstrate that the EFR signal provides incremental predictive power beyond closely related measures from the extensive anomalies literature. While share issuance and payout policies have been shown to predict returns ([Pontiff and Woodgate, 2008](#); [Boudoukh et al., 2007](#)), we establish that equity financing restraint captures distinct information about firms’ financial constraints and growth prospects that these existing measures miss. The signal maintains statistical and economic significance even after controlling for multiple related predictors, suggesting it identifies a unique dimension of firm behavior relevant for asset prices ([McLean and Pontiff, 2016](#)).

Third, we provide evidence that strengthens the information-based interpretation of cross-sectional return predictability by documenting how the EFR effect varies

systematically with proxies for information processing frictions. The concentration of returns among stocks with limited analyst coverage and lower institutional ownership directly supports models of limited attention and gradual information diffusion (Peng and Xiong, 2007; Hirshleifer et al., 2011). These findings have broader implications for understanding market efficiency and the role of information intermediaries in facilitating price discovery, particularly in segments of the market where professional attention is scarce.

2 Data

We obtain accounting data from the COMPUSTAT Annual database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and invested capital, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. construction of our Equity Financing Restraint signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of all common stock issued by the firm, as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the cumulative amount of common stock that has been issued to shareholders. Invested capital (ICAPT) measures the total capital invested in the firm’s operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable represents the permanent capital base that supports the firm’s operating activities. construct the Equity Financing Restraint signal by computing the year-over-year change in common stock, scaled by lagged invested capital: $(CSTK(t) - CSTK(t-1)) / ICAPT(t-1)$. This ratio captures the extent of new equity issuance activity relative to the firm’s existing capital base. A higher value indicates more aggressive equity

financing, while lower or negative values suggest restraint in accessing equity markets or potential share buyback activity. We calculate this measure using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. To maintain data quality, we require non-missing values for both CSTK and ICAPT in consecutive years, and we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme observations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFR signal. Panel A plots the time-series of the mean, median, and interquartile range for EFR. On average, the cross-sectional mean (median) EFR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFR signal for the CRSP universe. On average, the EFR signal is available for 6.37% of CRSP names, which on average make up 7.68% of total market capitalization.

4 Does EFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFR portfolio and sells the low EFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama

and French (2018) (FF6). The table shows that the long/short EFR strategy earns an average return of 0.31% per month with a t-statistic of 4.15. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.65 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 573 stocks and an average market capitalization of at least \$1,481 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.14. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-34bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.66. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the EFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFR, as well as average returns and alphas for long/short trading EFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EFR strategy achieves an average return of 24 bps/month with a t-statistic of 2.68. Among these large cap stocks, the alphas for the EFR strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.93 and 2.73.

5 How does EFR perform relative to the zoo?

Figure 2 puts the performance of EFR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EFR strategy falls in the distribution. The EFR strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFR strategy (red line).² Ignoring trading costs, a \$1 invested in the EFR strategy would have yielded \$7.45 which ranks the EFR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFR strategy would have yielded \$5.41 which ranks the EFR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFR relative to those. Panel A shows that the EFR strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFR ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFR or at least to weaken the power EFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR}EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFR signal in these Fama-MacBeth regressions exceed 2.83, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFR is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFR strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFR trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does EFR add relative to the whole zoo?

Finally, we can ask how much adding EFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFR grows to \$2676.66.

8 Conclusion

This study provides robust evidence that Equity Financing Restraint (EFR) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EFR maintains substantial predictive power even after accounting for transaction costs and controlling for established risk factors and related equity financing strategies.

The key findings reveal that a value-weighted long/short strategy based on EFR generates impressive risk-adjusted returns, with an annualized net Sharpe ratio of

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFR is available.

0.47 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (21 bps/month, $t\text{-stat} = 2.84$) and an expanded model including six closely related financing strategies (15 bps/month, $t\text{-stat} = 2.24$). The persistence of significant alpha even after controlling for similar equity issuance and payout-based factors underscores that EFR captures unique information about firm financing decisions and their implications for future returns.

From a practical perspective, these results suggest that EFR can serve as a valuable signal for portfolio construction and risk management. The strategy’s ability to generate positive risk-adjusted returns net of transaction costs indicates its potential implementability in real-world investment contexts. Moreover, the signal’s distinctiveness from existing equity financing measures highlights its incremental contribution to understanding the cross-section of expected returns.

However, several limitations warrant consideration. First, while our analysis accounts for standard transaction costs, actual implementation costs may vary depending on market conditions and trading infrastructure. Second, the study period and market coverage may influence the generalizability of results across different time periods and international markets. Third, the economic mechanisms underlying EFR’s predictive power deserve further theoretical and empirical investigation.

Future research should explore several promising directions. Examining the performance of EFR across different market regimes and economic cycles would enhance our understanding of its stability and conditional effectiveness. International evidence would help establish whether the signal’s predictive power extends beyond U.S. markets. Additionally, investigating the interaction between EFR and corporate governance, information asymmetry, or market sentiment could provide deeper insights into the economic channels through which financing constraints affect stock returns. Finally, combining EFR with machine learning techniques or alternative data sources may uncover non-linear relationships and enhance predictive accuracy.

Overall, this study establishes EFR as a robust addition to the factor zoo, while highlighting the continued importance of careful empirical validation in asset pricing research.

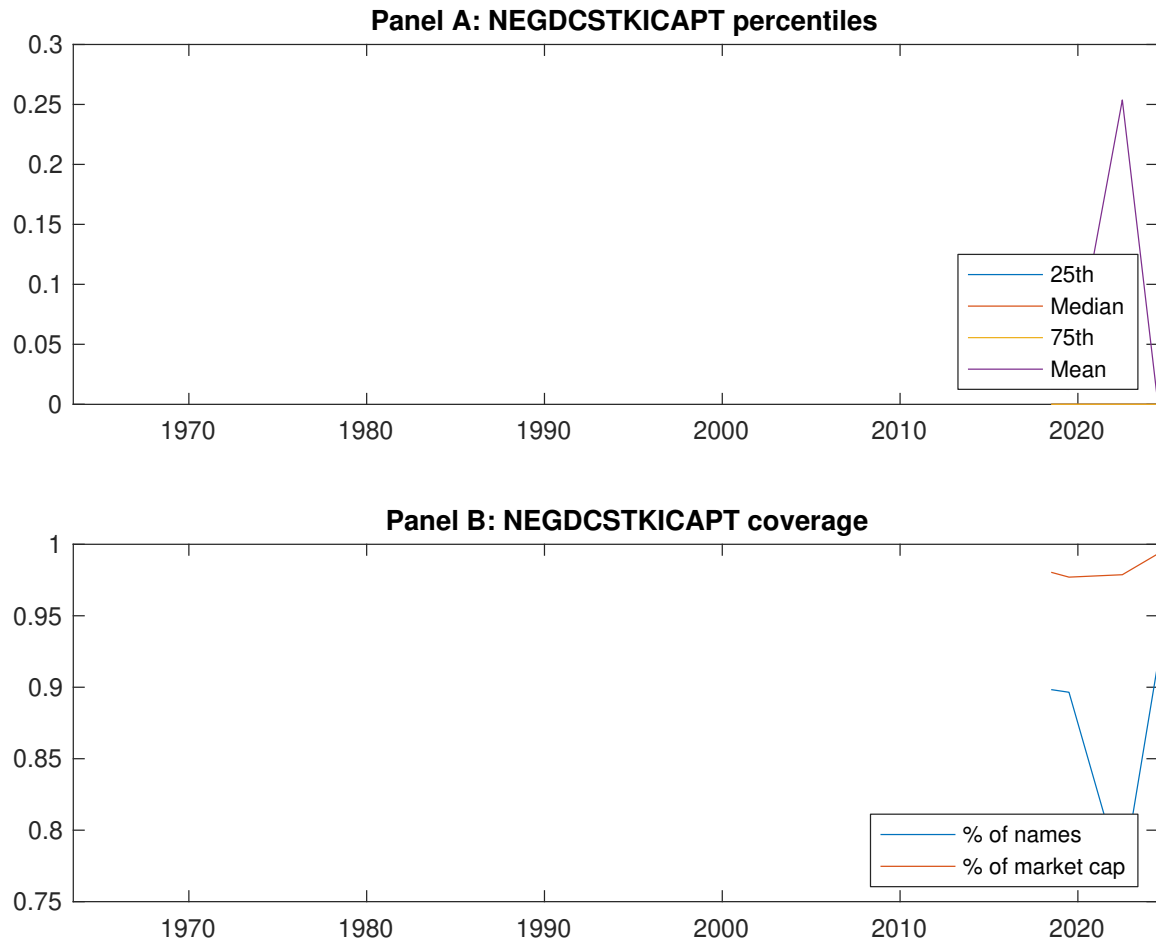


Figure 1: Times series of EFR percentiles and coverage.
This figure plots descriptive statistics for EFR. Panel A shows cross-sectional percentiles of EFR over the sample. Panel B plots the monthly coverage of EFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43 [2.56]	0.55 [3.05]	0.62 [3.47]	0.67 [4.10]	0.74 [4.68]	0.31 [4.15]
α_{CAPM}	-0.14 [-2.79]	-0.07 [-1.62]	0.01 [0.13]	0.11 [2.34]	0.20 [4.35]	0.34 [4.56]
α_{FF3}	-0.14 [-2.81]	-0.06 [-1.37]	0.02 [0.51]	0.07 [1.61]	0.16 [3.57]	0.30 [4.02]
α_{FF4}	-0.12 [-2.35]	-0.03 [-0.65]	0.04 [0.82]	0.03 [0.73]	0.15 [3.38]	0.27 [3.59]
α_{FF5}	-0.16 [-3.24]	0.00 [0.01]	0.04 [0.78]	0.01 [0.12]	0.07 [1.64]	0.24 [3.16]
α_{FF6}	-0.15 [-2.85]	0.02 [0.50]	0.05 [1.02]	-0.02 [-0.49]	0.07 [1.71]	0.22 [2.93]
Panel B: Fama and French (2018) 6-factor model loadings for EFR-sorted portfolios						
β_{MKT}	0.97 [78.87]	1.02 [94.38]	1.02 [88.97]	1.01 [91.17]	0.97 [93.02]	0.00 [0.01]
β_{SMB}	-0.02 [-0.86]	0.02 [1.42]	0.04 [2.53]	-0.07 [-4.55]	-0.01 [-0.42]	0.01 [0.34]
β_{HML}	0.05 [2.13]	-0.01 [-0.39]	-0.07 [-3.20]	0.08 [3.91]	0.02 [1.03]	-0.03 [-0.85]
β_{RMW}	0.13 [5.34]	-0.11 [-5.31]	-0.03 [-1.56]	0.10 [4.46]	0.13 [6.13]	-0.00 [-0.10]
β_{CMA}	-0.11 [-3.04]	-0.10 [-3.23]	0.01 [0.28]	0.15 [4.80]	0.24 [7.92]	0.34 [6.65]
β_{UMD}	-0.03 [-2.14]	-0.03 [-2.93]	-0.02 [-1.52]	0.04 [3.67]	-0.01 [-0.59]	0.02 [1.12]
Panel C: Average number of firms (n) and market capitalization (me)						
n	759	685	573	674	736	
me (\$10 ⁶)	1751	1481	2106	2276	2519	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.15]	0.34 [4.56]	0.30 [4.02]	0.27 [3.59]	0.24 [3.16]	0.22 [2.93]
Quintile	NYSE	EW	0.53 [8.77]	0.60 [10.30]	0.53 [9.67]	0.45 [8.39]	0.41 [7.82]	0.36 [6.87]
Quintile	Name	VW	0.31 [4.09]	0.33 [4.29]	0.30 [3.84]	0.27 [3.42]	0.24 [3.17]	0.23 [2.93]
Quintile	Cap	VW	0.24 [3.14]	0.26 [3.43]	0.23 [3.07]	0.19 [2.47]	0.21 [2.75]	0.18 [2.31]
Decile	NYSE	VW	0.26 [2.94]	0.28 [3.07]	0.21 [2.41]	0.17 [1.91]	0.21 [2.39]	0.18 [2.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.65]	0.30 [4.06]	0.26 [3.54]	0.25 [3.32]	0.22 [2.96]	0.21 [2.84]
Quintile	NYSE	EW	0.34 [5.00]	0.42 [6.44]	0.35 [5.70]	0.31 [5.12]	0.23 [4.05]	0.21 [3.67]
Quintile	Name	VW	0.27 [3.59]	0.29 [3.81]	0.25 [3.36]	0.24 [3.13]	0.22 [2.92]	0.21 [2.79]
Quintile	Cap	VW	0.20 [2.66]	0.22 [2.96]	0.19 [2.59]	0.17 [2.26]	0.18 [2.47]	0.17 [2.24]
Decile	NYSE	VW	0.22 [2.46]	0.23 [2.59]	0.17 [1.98]	0.15 [1.71]	0.18 [2.01]	0.16 [1.83]

Table 3: Conditional sort on size and EFR

This table presents results for conditional double sorts on size and EFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFR and short stocks with low EFR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EFR Quintiles					EFR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [1.92]	0.65 [2.57]	0.87 [3.46]	0.89 [3.62]	1.02 [4.46]	0.56 [7.07]	0.61 [7.75]	0.56 [7.33]	0.51 [6.56]	0.45 [5.92]	0.42 [5.38]
	(2)	0.56 [2.46]	0.71 [3.01]	0.90 [3.82]	0.89 [3.99]	0.96 [4.49]	0.41 [4.81]	0.46 [5.40]	0.38 [4.62]	0.33 [3.95]	0.32 [3.90]	0.29 [3.43]
	(3)	0.56 [2.78]	0.64 [3.00]	0.76 [3.46]	0.84 [4.14]	0.91 [4.70]	0.35 [4.76]	0.38 [5.04]	0.33 [4.45]	0.29 [3.87]	0.29 [3.86]	0.26 [3.44]
	(4)	0.51 [2.68]	0.61 [3.10]	0.78 [3.84]	0.77 [4.12]	0.81 [4.50]	0.30 [4.03]	0.34 [4.53]	0.27 [3.83]	0.26 [3.59]	0.13 [1.84]	0.13 [1.86]
	(5)	0.45 [2.74]	0.48 [2.66]	0.50 [2.89]	0.56 [3.41]	0.69 [4.36]	0.24 [2.68]	0.25 [2.73]	0.22 [2.41]	0.18 [1.93]	0.21 [2.28]	0.18 [1.93]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EFR Quintiles					EFR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	382	383	382	381	374	31	33	41	30	29	
	(2)	109	107	108	108	106	57	56	58	56	56	
	(3)	78	78	77	77	78	98	95	98	100	100	
	(4)	65	65	65	65	66	204	205	216	212	218	
(5)	60	60	60	60	60	1478	1429	1664	1677	1898		

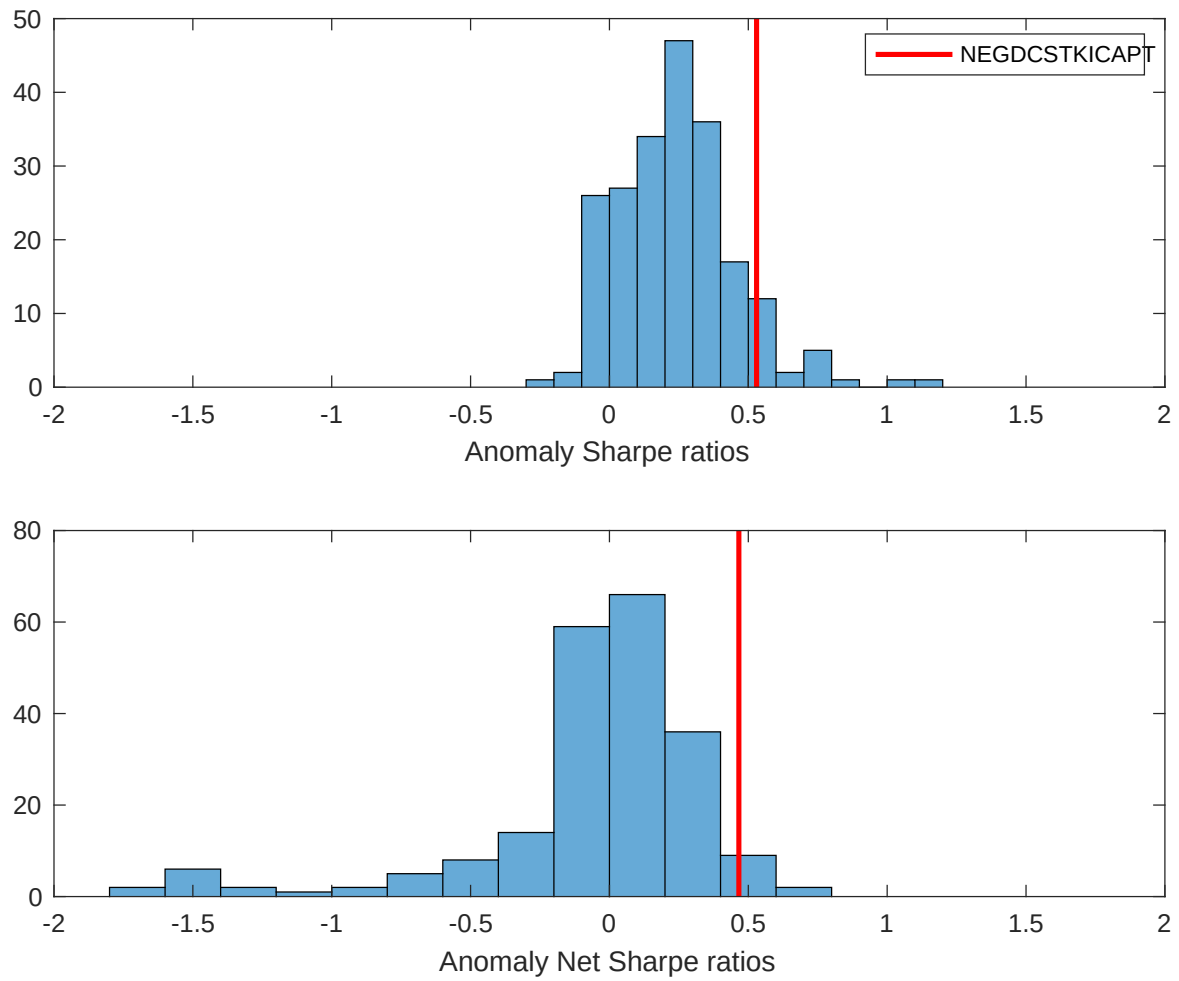


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

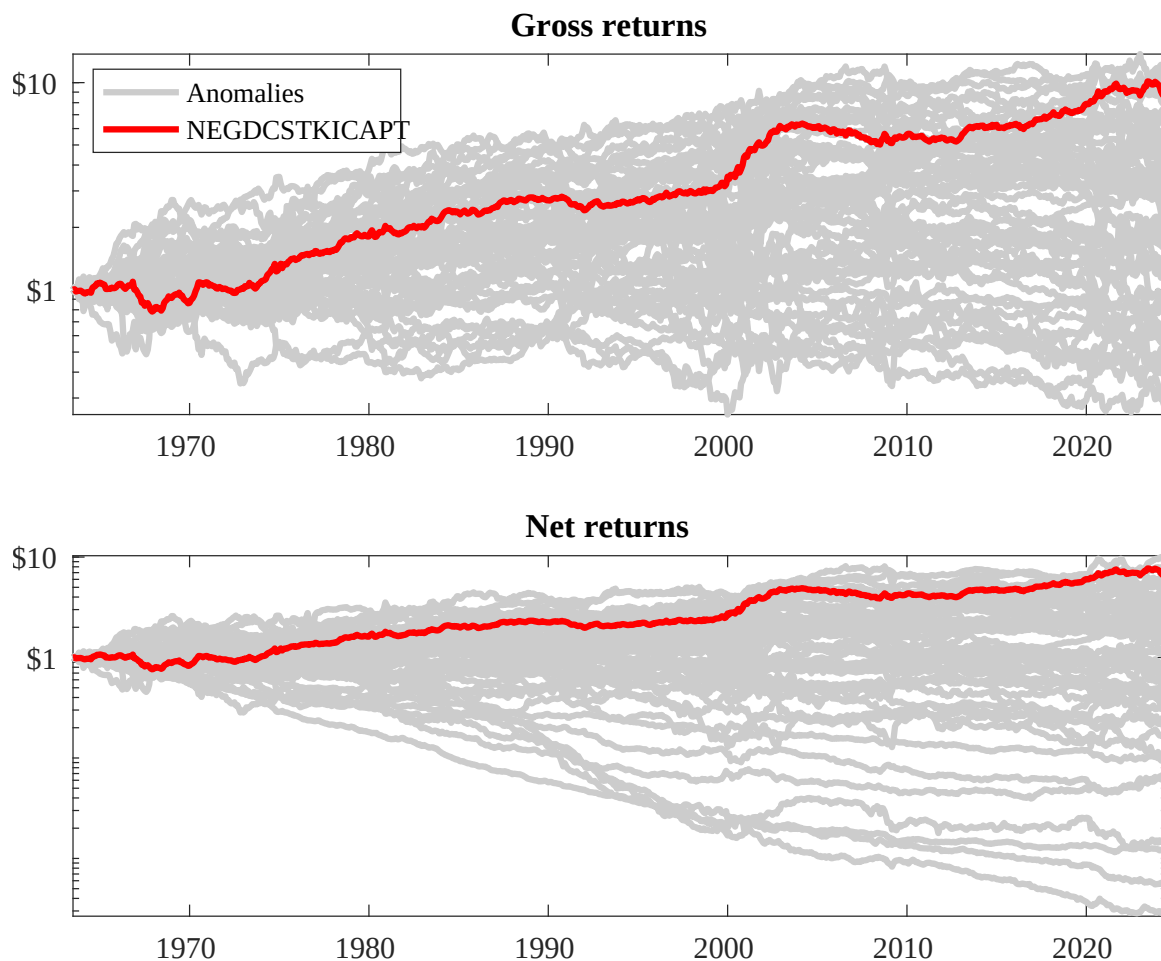
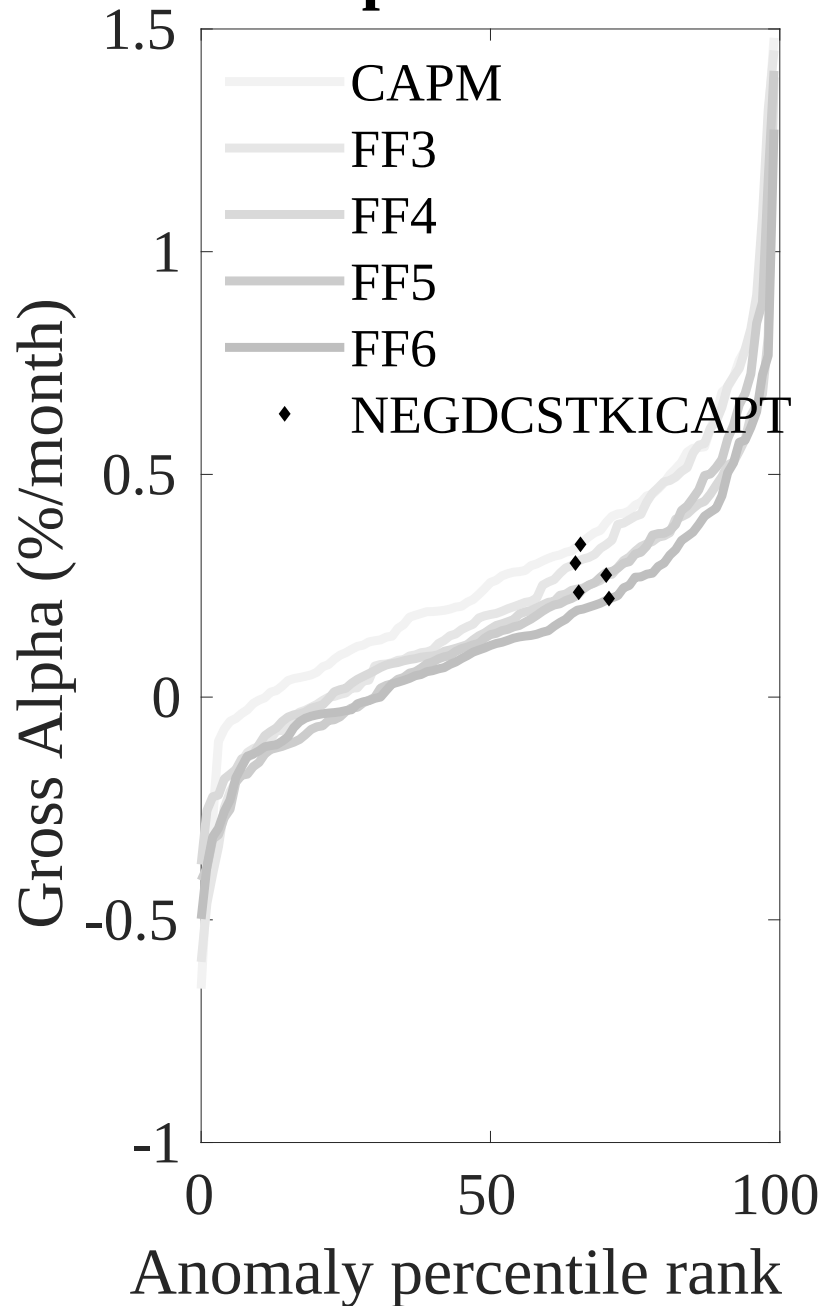


Figure 3: Dollar invested.

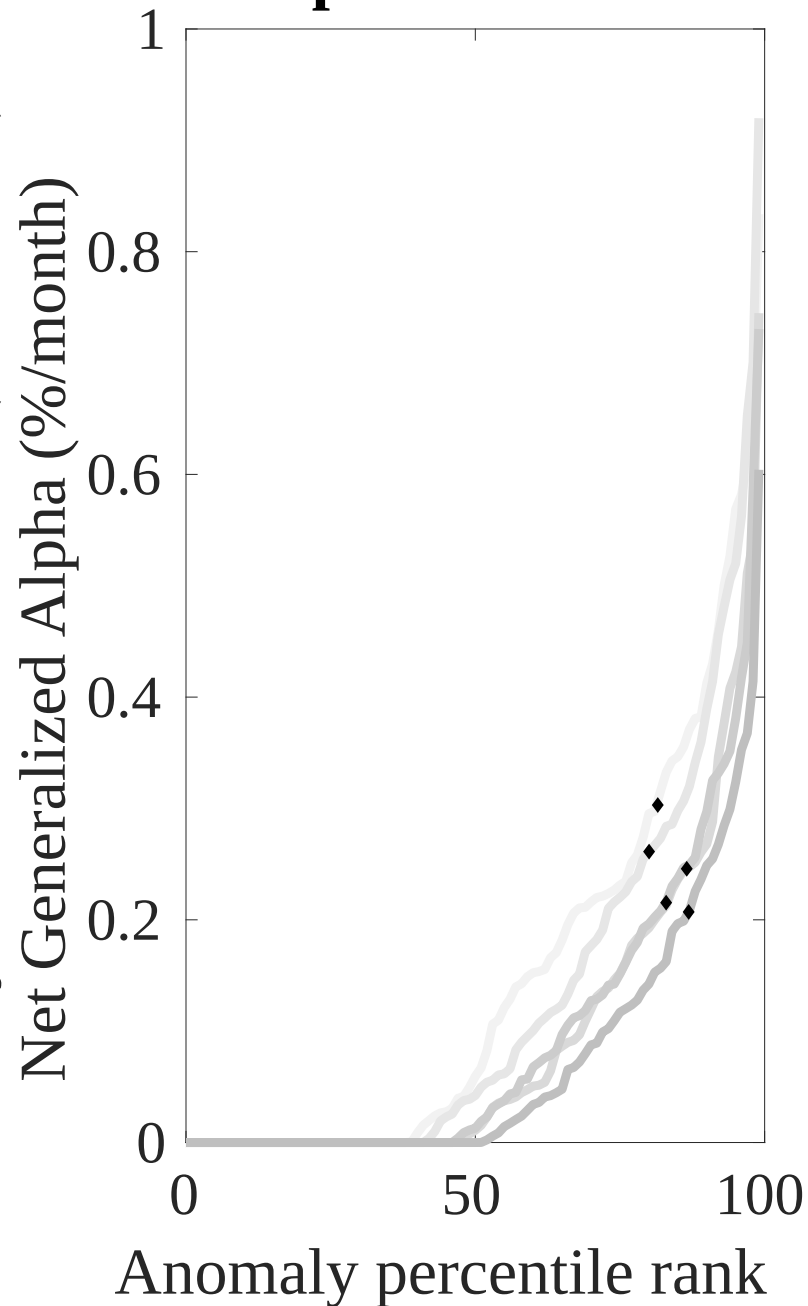
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



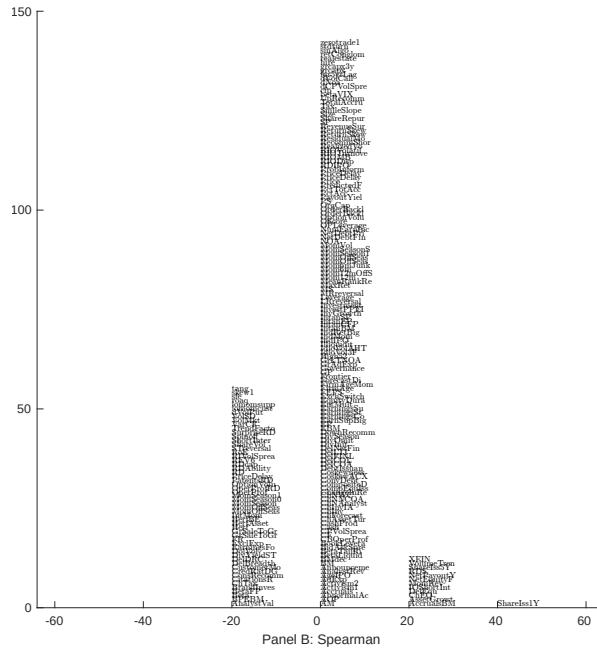
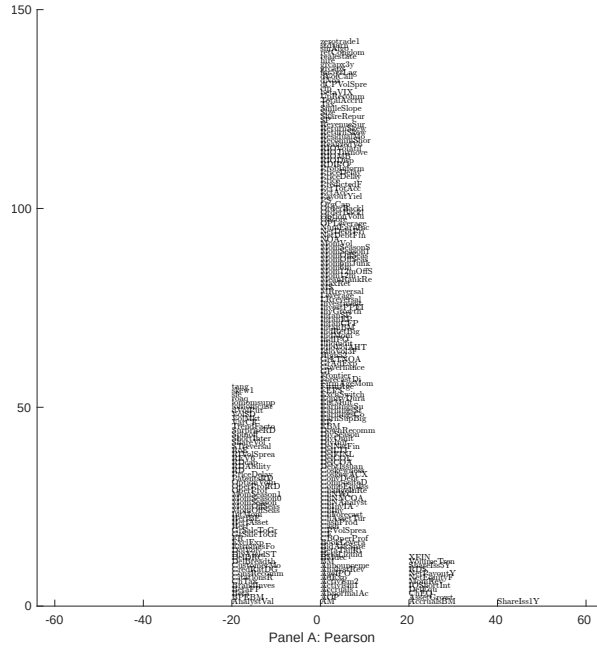


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

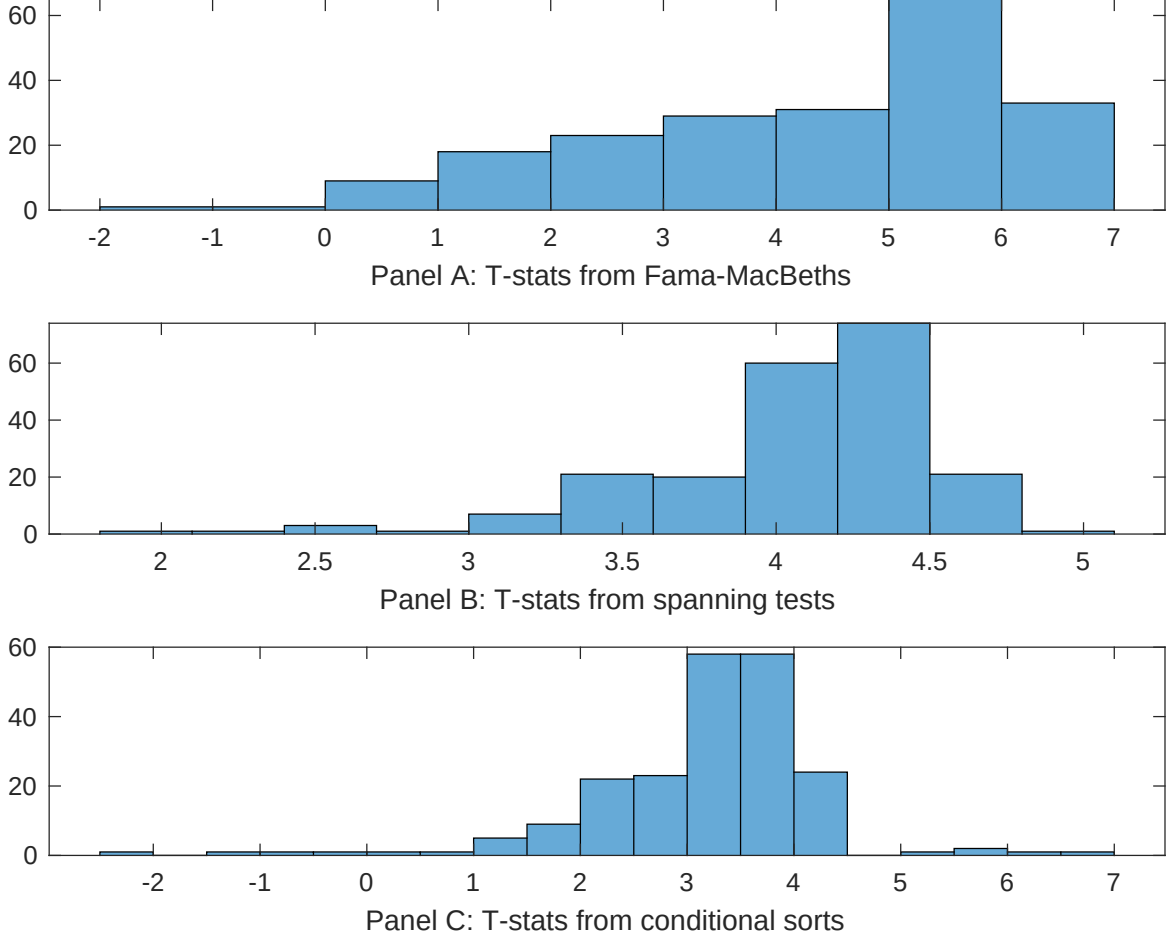


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.16 [6.82]	0.13 [6.45]	0.12 [5.43]	0.12 [5.71]	0.13 [6.17]	0.12 [3.91]
EFR	0.40 [4.82]	0.36 [3.88]	0.36 [4.39]	0.30 [2.83]	0.44 [4.82]	0.37 [3.99]	0.15 [1.47]
Anomaly 1	0.13 [5.87]						0.66 [2.40]
Anomaly 2		0.37 [3.17]					-0.13 [-0.59]
Anomaly 3			0.26 [4.77]				0.24 [4.27]
Anomaly 4				0.21 [1.85]			0.93 [0.88]
Anomaly 5					0.11 [3.08]		-0.64 [-0.11]
Anomaly 6						0.92 [7.86]	0.65 [5.45]
# months	715	720	715	715	720	720	715
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.76]	0.20 [2.83]	0.15 [1.99]	0.25 [3.36]	0.22 [2.94]	0.21 [2.85]	0.15 [2.24]
Anomaly 1	30.08 [8.25]						15.75 [3.61]
Anomaly 2		31.95 [8.21]					37.03 [6.67]
Anomaly 3			28.73 [7.40]				12.97 [3.00]
Anomaly 4				17.51 [6.16]			5.57 [1.83]
Anomaly 5					13.68 [3.70]		-9.63 [-1.89]
Anomaly 6						-3.55 [-0.80]	-21.73 [-4.55]
mkt	0.93 [0.54]	1.25 [0.73]	1.48 [0.85]	2.91 [1.62]	-0.05 [-0.03]	0.20 [0.11]	3.33 [1.96]
smb	3.19 [1.29]	0.93 [0.38]	3.91 [1.55]	4.84 [1.88]	1.83 [0.71]	2.29 [0.88]	5.15 [2.10]
hml	-2.68 [-0.81]	-5.72 [-1.71]	0.44 [0.13]	-8.67 [-2.42]	-3.59 [-1.04]	-1.61 [-0.46]	-6.15 [-1.82]
rmw	-16.68 [-4.29]	1.18 [0.35]	-10.68 [-2.92]	-10.58 [-2.79]	0.92 [0.26]	-0.28 [-0.08]	-15.74 [-3.90]
cma	14.90 [2.79]	2.53 [0.41]	19.63 [3.75]	18.81 [3.43]	19.71 [3.15]	37.77 [5.09]	13.12 [1.80]
umd	0.81 [0.47]	1.62 [0.95]	0.30 [0.18]	3.58 [2.05]	2.52 [1.43]	2.11 [1.18]	-0.51 [-0.30]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	18	17	16	14	11	10	26

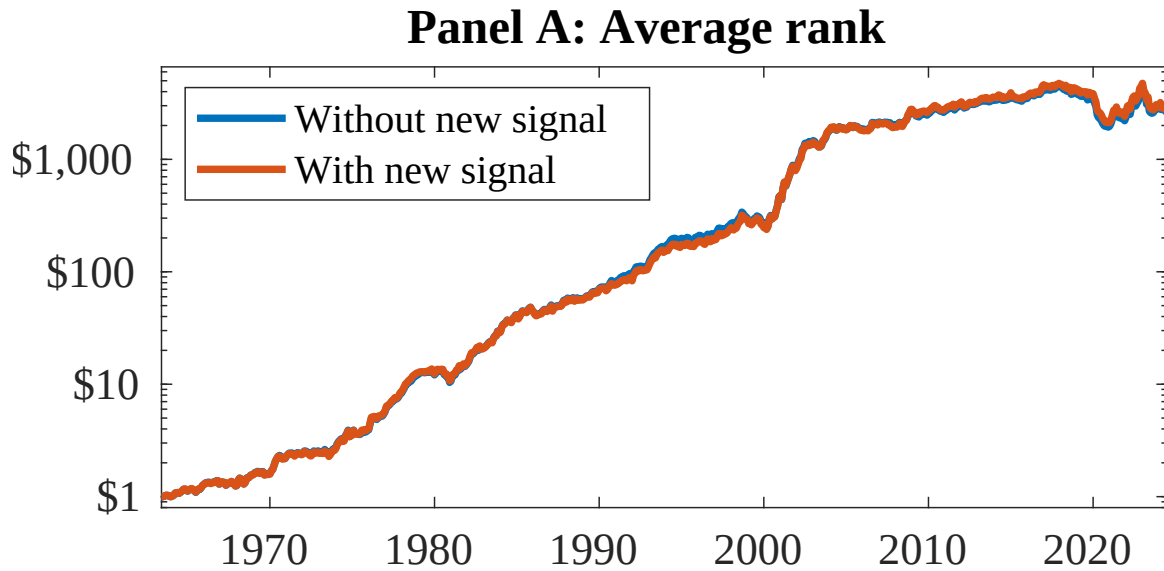


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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